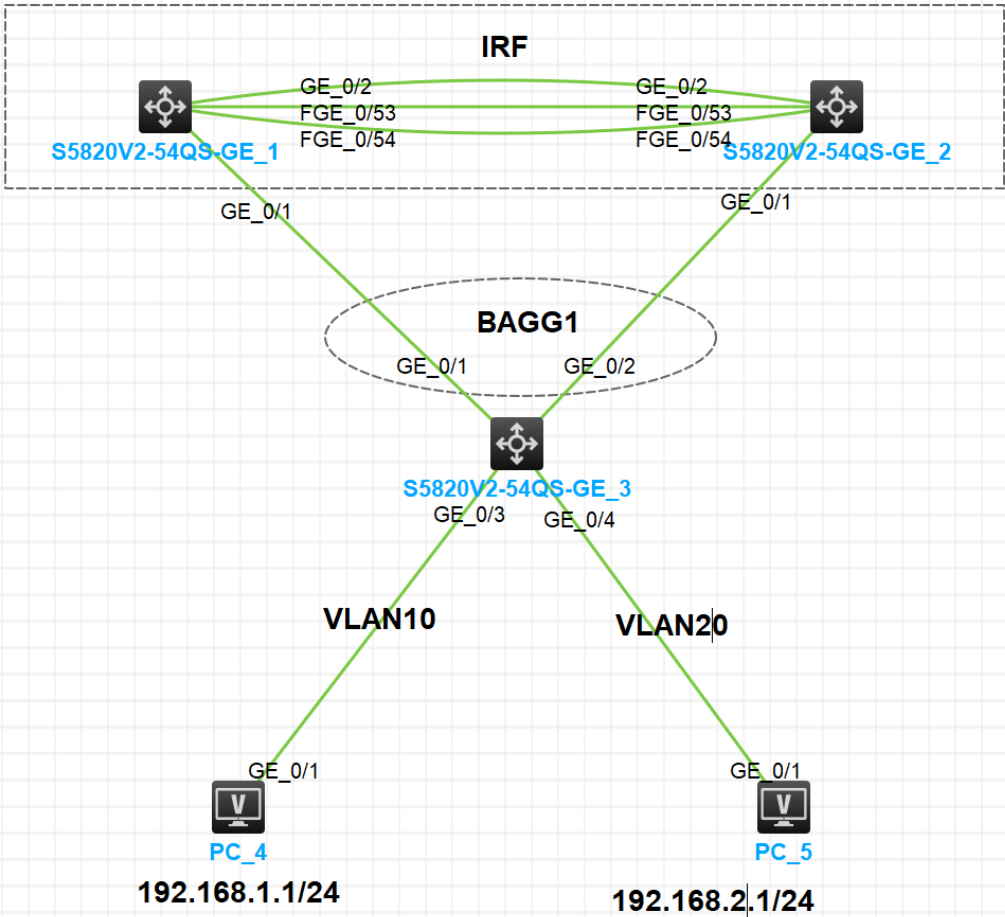


IRF与MAD实验

实验拓扑

- 实验需求
- 1. SW1 和 SW2 配置 IRF，作为核心交换机与 SW3 作为接入交换机对接
 - 2. SW3 作为接入交换机把连接 PC4 的接口加入 VLAN 10，连接 PC5 的接口加入 VLAN 20。
 - SW3 与核心的堆叠设备配置跨设备链路聚合
 - 3. 在堆叠设备上配置 BFD MAD 检测防止堆叠分裂



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实验配置

```

[SW1-if-range]int range f1/0/53 to f1/0/54
[SW1-if-range]shutdown
[SW1-if-range]qu
[SW1]irf-port 1/1
[SW1-irf-port1/1]port group interface FortyGigE 1/0/53
You must perform the following tasks for a successful IRF setup:
Save the configuration after completing IRF configuration.
Execute the "irf-port-configuration active" command to activate the IRF ports.
[SW1-irf-port1/1]port group interface FortyGigE 1/0/54
[SW1]int ra f1/0/53 to f1/0/54
[SW1-if-range]un shutdown
<SW1>sa
[SW1]irf-port-configuration active
[SW1]interface Bridge-Aggregation 1
[IRF]int ra g1/0/1 g2/0/1
[IRF-if-range]port link-aggregation group 1
[IRF]vlan 10
[IRF-vlan10]vlan 20
[SW1]interface Bridge-Aggregation 1
[IRF-Bridge-Aggregation1]port link-type trunk
Please wait.....Done.
Configuring GigabitEthernet1/0/1 done.
Configuring GigabitEthernet2/0/1 done.
[IRF-Bridge-Aggregation1]port trunk permit vlan 10 20
Please wait.....Done.
Configuring GigabitEthernet1/0/1 done.
Configuring GigabitEthernet2/0/1 done.
[IRF-if-range]vlan 2
[IRF]interface vlan 2
[IRF-Vlan-interface2]mad bfd enable
[IRF-Vlan-interface2]mad ip address 10.1.1.1 24 member 1
[IRF-Vlan-interface2]mad ip address 10.1.1.2 24 member 2
[IRF-Vlan-interface2]int ra g1/0/2 g2/0/2
[IRF-if-range]un stp en
---
[SW2]irf member 1 renumber 2
Renumbering the member ID may result in configuration change or loss. Continue?[Y/N]:y
<SW2>sa
<SW2>reboot
Start to check configuration with next startup configuration file, please wait.....DONE!
This command will reboot the device. Continue? [Y/N]:y
[SW2]dis interface brief

```

Interface	Link	Speed	Duplex	Type	PVID	Description
FGE2/0/53	UP	40G	F(a)	A	1	
FGE2/0/54	UP	40G	F(a)	A	1	
GE2/0/1	UP	1G(a)	F(a)	A	1	
GE2/0/2	UP	1G(a)	F(a)	A	1	

```

[SW2]int ra f2/0/53 to f 2/0/54
[SW2-if-range]shutdown
[SW2]irf-port 2/2
[SW2-irf-port2/2]port group interface FortyGigE 2/0/53

```

```

You must perform the following tasks for a successful IRF setup:
Save the configuration after completing IRF configuration.
Execute the "irf-port-configuration active" command to activate the IRF ports.
[SW2-irf-port2/2]port group interface FortyGigE 2/0/54
[SW2]int ra FortyGigE 2/0/53 to f 2/0/54
[SW2-if-range]un shutdown
<SW2>sa
[SW2]irf-port-configuration active
---
[SW3]vlan 10
[SW3-vlan10]vlan 20
[SW3-vlan20]int g1/0/3
[SW3-GigabitEthernet1/0/3]port link-type access
[SW3-GigabitEthernet1/0/3]port access vlan 10
[SW3-GigabitEthernet1/0/3]int g1/0/4
[SW3-GigabitEthernet1/0/4]port link-type access
[SW3-GigabitEthernet1/0/4]port access vlan 20
[SW3]interface Bridge-Aggregation 1
[SW3]int ra g1/0/1 to g1/0/2
[SW3-if-range]port link-aggregation group 1
[SW3]int Bridge-Aggregation 1
[SW3-Bridge-Aggregation1]port link-type trunk
Configuring GigabitEthernet1/0/1 done.
Configuring GigabitEthernet1/0/2 done.
[SW3-Bridge-Aggregation1]port trunk permit vlan 10 20
Configuring GigabitEthernet1/0/1 done.
Configuring GigabitEthernet1/0/2 done.

```

结果验证

1.查看IRF配置信息

```

[IRF]display irf
MemberID    Role    Priority  CPU-Mac          Description
*+1         Master  1         7455-6e91-0104   ---
  2         Standby 1         7455-6fb5-0204   ---
-----
* indicates the device is the master.
+ indicates the device through which the user logs in.

The bridge MAC of the IRF is: 7455-6e91-0100
Auto upgrade           : yes
Mac persistent         : 6 min
Domain ID              : 0

```

2.查看链路聚合组

```
[IRF]display link-aggregation summary
Aggregation Interface Type:
BAGG -- Bridge-Aggregation, BLAGG -- Blade-Aggregation, RAGG -- Route-Aggregation, SCH-B --
Schannel-Bundle
Aggregation Mode: S -- Static, D -- Dynamic
Loadsharing Type: Shar -- Loadsharing, NonS -- Non-Loadsharing
Actor System ID: 0x8000, 7455-6e91-0100
```

AGG Interface	AGG Mode	Partner ID	Selected Ports	Unselected Ports	Individual Ports	Share Type
BAGG1	S	None	2	0	0	Shar

```
[IRF]display link-aggregation b
[IRF]display link-aggregation ?
  load-sharing  Link aggregation load sharing
  member-port   Display member(s) of link aggregation group
  summary       Summary information
  verbose       Display verbose information of link aggregation group
```

```
[IRF]display link-aggregation verbose
Loadsharing Type: Shar -- Loadsharing, NonS -- Non-Loadsharing
Port: A -- Auto
Port Status: S -- Selected, U -- Unselected, I -- Individual
Flags:  A -- LACP_Activity, B -- LACP_Timeout, C -- Aggregation,
        D -- Synchronization, E -- Collecting, F -- Distributing,
        G -- Defaulted, H -- Expired
```

```
Aggregate Interface: Bridge-Aggregation1
Aggregation Mode: Static
Loadsharing Type: Shar
  Port          Status  Priority Oper-Key
```

GE1/0/1	S	32768	1
GE2/0/1	S	32768	1

```
<SW3>display link-aggregation verbose
Loadsharing Type: Shar -- Loadsharing, NonS -- Non-Loadsharing
Port: A -- Auto
Port Status: S -- Selected, U -- Unselected, I -- Individual
Flags:  A -- LACP_Activity, B -- LACP_Timeout, C -- Aggregation,
        D -- Synchronization, E -- Collecting, F -- Distributing,
        G -- Defaulted, H -- Expired
```

```
Aggregate Interface: Bridge-Aggregation1
Aggregation Mode: Static
Loadsharing Type: Shar
  Port          Status  Priority Oper-Key
```

GE1/0/1	S	32768	1
GE1/0/2	S	32768	1

```
<SW3>display link-aggregation summary
Aggregation Interface Type:
BAGG -- Bridge-Aggregation, BLAGG -- Blade-Aggregation, RAGG -- Route-Aggregation, SCH-B --
Schannel-Bundle
Aggregation Mode: S -- Static, D -- Dynamic
Loadsharing Type: Shar -- Loadsharing, NonS -- Non-Loadsharing
Actor System ID: 0x8000, 7455-7d02-0300
```

AGG Interface	AGG Mode	Partner ID	Selected Ports	Unselected Ports	Individual Ports	Share Type
BAGG1	S	None	2	0	0	Shar

3.查看VLAN信息

```
[IRF]display vlan brief
Brief information about all VLANs:
Supported Minimum VLAN ID: 1
Supported Maximum VLAN ID: 4094
Default VLAN ID: 1
```

VLAN ID	Name	Port
1	VLAN 0001	BAGG1 GE1/0/1 GE1/0/3 GE1/0/4 GE1/0/5 GE1/0/6 GE1/0/7 GE1/0/8 GE1/0/9 GE1/0/10 GE1/0/11 GE1/0/12 GE1/0/13 GE1/0/14 GE1/0/15 GE1/0/16 GE1/0/17 GE1/0/18 GE1/0/19 GE1/0/20
2	VLAN 0002	GE1/0/2 GE2/0/2
10	VLAN 0010	BAGG1 GE1/0/1 GE2/0/1
20	VLAN 0020	BAGG1 GE1/0/1 GE2/0/1

```
<SW3>display vlan brief
Brief information about all VLANs:
Supported Minimum VLAN ID: 1
Supported Maximum VLAN ID: 4094
Default VLAN ID: 1
```

VLAN ID	Name	Port
1	VLAN 0001	BAGG1 FGE1/0/53 FGE1/0/54 GE1/0/1 GE1/0/2 GE1/0/5 GE1/0/6 GE1/0/7 GE1/0/8 GE1/0/9 GE1/0/10 GE1/0/11 GE1/0/12 GE1/0/13 GE1/0/14 GE1/0/15 GE1/0/16 GE1/0/17 GE1/0/18 GE1/0/19 GE1/0/20 GE1/0/21 GE1/0/22 GE1/0/23 GE1/0/24 GE1/0/25 GE1/0/26 GE1/0/27 GE1/0/28 GE1/0/29 GE1/0/30 GE1/0/31 GE1/0/32 GE1/0/33 GE1/0/34 GE1/0/35 GE1/0/36 GE1/0/37 GE1/0/38 GE1/0/39 GE1/0/40 GE1/0/41 GE1/0/42 GE1/0/43

		GE1/0/44	GE1/0/45	GE1/0/46	
		GE1/0/47	GE1/0/48	XGE1/0/49	
		XGE1/0/50	XGE1/0/51	XGE1/0/52	
10	VLAN 0010	BAGG1	GE1/0/1	GE1/0/2	GE1/0/3
20	VLAN 0020	BAGG1	GE1/0/1	GE1/0/2	GE1/0/4

4.查看BFD会话信息

```
[IRF]display bfd session verbose
Total Session Num: 1      Up Session Num: 0      Init Mode: Active

IPv4 session working in control packet mode:
    Local Discr: 129          Remote Discr: 0
    Source IP: 10.1.1.1      Destination IP: 10.1.1.2
Session State: Down          Interface: Vlan-interface2
Min Tx Inter: 400ms          Act Tx Inter: 100ms
Min Rx Inter: 400ms          Detect Inter: 5000ms
Rx Count: 0                  Tx Count: 13868
Connect Type: Direct          Running Up for: 00:00:00
Hold Time: 0ms               Auth mode: None
Detect Mode: Async           Slot: 1
Protocol: MAD
Version: 1
Diag Info: No Diagnostic
```

5.查看接口信息

```
---
[IRF]dis int brief
Brief information on interfaces in route mode:
Link: ADM - administratively down; Stby - standby
Protocol: (s) - spoofing
Interface      Link Protocol Primary IP      Description
InLoop0        UP    UP(s)    --
MGE0/0/0        DOWN DOWN    --
NULL0          UP    UP(s)    --
REG0           UP    --       --
Vlan2          UP    UP       10.1.1.1

Brief information on interfaces in bridge mode:
Link: ADM - administratively down; Stby - standby
Speed: (a) - auto
Duplex: (a)/A - auto; H - half; F - full
Type: A - access; T - trunk; H - hybrid
Interface      Link Speed  Duplex Type PVID Description
BAGG1          UP    2G(a)    F(a)   A    1
FGE1/0/53      UP    40G      F(a)   --   --
FGE1/0/54      UP    40G      F(a)   --   --
FGE2/0/53      UP    40G      F(a)   --   --
FGE2/0/54      UP    40G      F(a)   --   --
```

附录

1.在SW3上配置链路聚合组时发现未成功

在SW3上的聚合组1中使用dis th命令，发现未配置trunk和放行VLAN10、20；而在1/0/1和1/0/2接口下发现配置了trunk和放行了VLAN10、20；

解决：取消接口下的trunk和放行VLAN 10、20；在聚合组中配置trunk和放行VLAN10、20